

Lesson 11. Partial Differential Equations III

University of Budapest

Faculty of Informatics, Department of Numerical Analysis

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The Laplacian in Polar Coordinates

Why Polar Coordinates?

- Polar coordinates simplify solving problems in **circular domains**.
- The boundary of a circle is easily expressed as $r = R$, removing complexity in boundary conditions.

The Laplacian in Polar Form

In polar coordinates (r, θ) , the Laplacian operator becomes:

$$\Delta u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}.$$

- The first two terms handle radial variations.
- The third term accounts for angular variations.

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The Laplacian in Cylindrical Coordinates

Why Cylindrical Coordinates?

- Cylindrical coordinates are ideal for solving problems in **cylindrical domains**, such as pipes, rods, or cylinders.
- They simplify problems with boundaries that are symmetric around an axis.

The Laplacian in Cylindrical Form

In cylindrical coordinates (r, θ, z) , the Laplacian operator becomes:

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- First two terms: Radial variations.
- Third term: Angular variations.
- Fourth term: Variations along the z -axis.

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The Laplacian in Spherical Coordinates

Why Spherical Coordinates?

- Spherical coordinates are ideal for solving problems in **spherical domains**, such as spheres or regions with radial symmetry.
- These coordinates simplify problems where boundaries are described by constant radius values.

The Laplacian in Spherical Form

In spherical coordinates (r, θ, ϕ) , the Laplacian operator becomes:

$$\Delta u = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2}.$$

- First term: Radial variation.
- Second term: Polar (angular) variation.
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Bessel Functions

Definition

Bessel functions are solutions to the *Bessel's differential equation*.

Importance

The two most commonly used types are

- the Bessel functions of the first kind $J_n(x)$;
- the Bessel functions of the second kind $Y_n(x)$.

Formula for Bessel Functions of the First Kind

The Bessel function of the first kind of order n is given by the following series representation:

$$J_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2}\right)^{2m+n}$$

where x is the argument and n is the order of the Bessel function.

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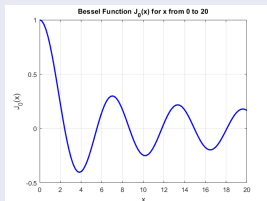
Bessel Functions

For example, for $n = 0$, the formula becomes:

$$J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2}\right)^{2m}$$

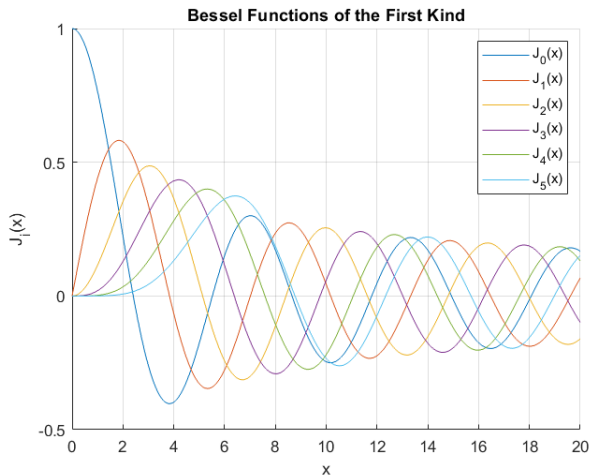
Visualization of Bessel Functions

The Bessel functions $J_0(x)$ and $J_1(x)$ oscillate and decay as x increases. Below is the plot of $J_0(x)$ for x from 0 to 20.



As you can see, $J_0(x)$ starts at 1 for $x = 0$, then oscillates and decays.

Bessel Functions



Bessel Functions

Bessel Function of the Second Kind $Y_n(x)$

- The Bessel function of the second kind $Y_n(x)$ can be expressed as:

$$Y_n(x) = \frac{J_n(x) \cos(n\pi) - J_{-n}(x)}{\sin(n\pi)}$$

- Alternatively, $Y_n(x)$ can be computed using the following series expansion:

$$Y_n(x) = \frac{2}{\pi} \left[J_n(x) \ln \left(\frac{x}{2} \right) - \sum_{m=1}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2} \right)^{2m+n} \right]$$

Vibrations of a Circular Membrane: Symmetric Case

Introduction to the Problem

- We are studying the vibrations of a thin circular membrane with uniform mass density, clamped along its circumference.
- The membrane's center is located at the origin, and its radius is denoted by a .
- The vibrations of the membrane are governed by the **two-dimensional wave equation**, which describes how the membrane's displacement evolves over time.
- Since the problem has radial symmetry, we express the wave equation in **polar coordinates**, which are better suited for this problem due to the circular geometry of the membrane.

Initial Conditions

- The initial shape of the membrane is modeled by the function $f(r, \theta)$, which describes the displacement at time $t = 0$;
- The initial velocity of the membrane is given by the function $g(r, \theta)$, which represents the rate of change of displacement with respect to time at $t = 0$.

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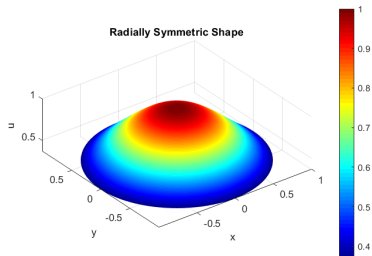
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Vibrations of a Circular Membrane: Symmetric Case

Axisymmetric Assumption

- In this lesson, we focus on the case where both the initial displacement $f(r, \theta)$ and the initial velocity $g(r, \theta)$ are radially symmetric (also called axisymmetric). This means that both functions depend only on the radial coordinate r , and not on the angular coordinate.
- Under this assumption, it is reasonable to expect that the solution to the wave equation will also be independent of θ , simplifying the problem and allowing us to focus on the radial behavior.



Vibrations of a Circular Membrane: Symmetric Case

Equation

The vibration equation for a thin circular membrane, where the solution does not depend on θ , becomes:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right), \quad (1)$$

where $u = u(r, t)$, $0 < r < a$, and $t > 0$.

Boundary conditions

Since the membrane is clamped at the circumference, we have the boundary condition:

$$u(a, t) = 0, \quad t \geq 0. \quad (2)$$

Initial conditions

The radially symmetric initial conditions are:

$$u(r, 0) = f(r), \quad \frac{\partial u}{\partial t}(r, 0) = g(r), \quad 0 < r < a. \quad (3)$$

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Vibrations of a Circular Membrane: Symmetric Case

We solve the boundary value problem (1)-(3) using the separation of variables method. The goal is to separate the variables in the partial differential equation (1) and reduce the problem to two ordinary differential equations.

Separating Variables

We assume that the solution is of the form $u(r, t) = R(r)T(t)$. After differentiating, plugging into (2), and separating variables, we get

$$\frac{T''}{c^2 T} = \frac{1}{R} (rR'' + R') = -\lambda^2.$$

Because we expect periodic solutions in T , we have set the sign of the separation constant negative. Thuse, we get

$$rr'' + r' + \lambda^2 rR = 0, \quad R(a) = 0 \quad (4)$$

$$T'' + c^2 \lambda^2 T = 0. \quad (5)$$

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Vibrations of a Circular Membrane: Symmetric Case

Solving the Separated Equations

Here again, we begin by solving the equation with the boundary conditions to narrow down the possible solutions. Equation (4) is known as the *parametric form of Bessel's equation of order zero* (here λ is the parameter).

This equation arises so frequently in applications that its solutions have been named. Since the equation is second order and homogeneous, we need only two linearly independent solutions to be able to write its general solution. These two linearly independent solutions are *Bessel functions of order 0 of the first and second kind*, $J_0(\lambda r)$ and $Y_0(\lambda r)$, respectively. Thus, the general solution to the parametric form of Bessel's equation in (4) is

$$R(r) = c_1 J_0(\lambda r) + c_2 Y_0(\lambda r). \quad (6)$$

Vibrations of a Circular Membrane: Symmetric Case

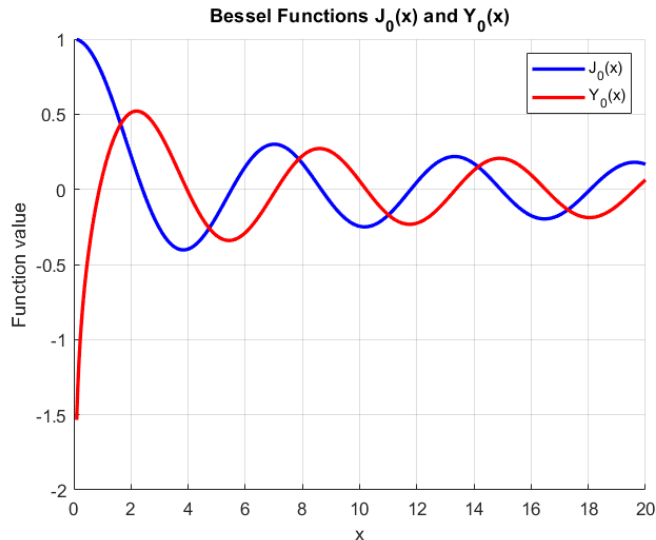
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Vibrations of a Circular Membrane: Symmetric Case



Vibrations of a Circular Membrane: Symmetric Case

Since the solutions to the wave equation are expected to be bounded, it follows that the spatial part of the solution, $R(r)$, has to be bounded near $r = 0$. This is effectively a second boundary condition on R .

Now the fact that Y_0 is unbounded near 0 forces us to choose $c_2 = 0$ in (6). To avoid trivial solutions, we will take $c_1 = 1$ and get

$$R(r) = J_0(\lambda r) \tag{7}$$

The condition $R(a) = 0$ implies that

$$J_0(\lambda a) = 0,$$

and so a must be a root of the Bessel function J_0 .

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Vibrations of a Circular Membrane: Symmetric Case

J_0 has infinitely many positive zeros, which we denote by

$$\alpha_1 < \alpha_2 < \dots < \alpha_n$$

Thus,

$$\lambda = \lambda_n = \frac{\alpha_n}{a} r$$

and the corresponding solutions of (4) are

$$R_n(r) = J_0\left(\frac{\alpha_n}{a} r\right)$$

where a_n is the n th positive zero of J_0 .

Returning to (5) with $\lambda = \lambda_n$, we find

$$T(t) = T_n(t) = A_n \cos(c\lambda_n t) + B_n \sin(c\lambda_n t)$$

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Vibrations of a Circular Membrane: Symmetric Case

We thus obtain the product solutions

$$u_n(r, t) = (A_n \cos(c\lambda_n t) + B_n \sin(c\lambda_n t))J_0(\lambda_n r)$$

To satisfy the initial conditions, motivated by the superposition principle, we let

$$u(r, t) = \sum_{n=1}^{\infty} (A_n \cos(c\lambda_n t) + B_n \sin(c\lambda_n t))J_0(\lambda_n r).$$

We determine the unknown coefficients by evaluating the series at $t = 0$ and using the initial conditions. We get from the first condition in (3):

$$u(r, 0) = f(r) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r), \quad 0 < r < a$$

The series expansions that arise are known as Bessel, or Fourier-Bessel, expansions.

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Vibrations of a Circular Membrane: Symmetric Case

WAVE EQUATION IN POLAR COORDINATES

The solution of the radially symmetric two-dimensional wave equation (1) with boundary and initial conditions (2) and (3) is

$$u(r, t) = \sum_{n=1}^{\infty} (A_n \cos(c\lambda_n t) + B_n \sin(c\lambda_n t)) J_0(\lambda_n r) \quad (8)$$

where

$$A_n = \frac{2}{a^2 J_1^2(\alpha_n)} \int_0^a f(r) J_0(\lambda_n r) r dr \quad (9)$$

$$B_n = \frac{2}{c\alpha_n a J_1^2(\alpha_n)} \int_0^a g(r) J_0(\lambda_n r) r dr \quad (10)$$

$\lambda_n = \frac{\alpha_n}{a}$, $n = 1, 2, 3, \dots$ and α_n is the n th positive zero of J_0 .

Vibrations of a Circular Membrane: Symmetric Case

Example: circular membrane with constant initial velocity

An explosion near the surface of a flexible circular membrane with clamped edges imparts a uniform initial velocity equal to -100 m/sec . Assume the initial shape of the membrane to be flat, take $a = 1$ and $c = 100$, and determine the subsequent vibrations of the membrane.

Solution

The solution is given by (8)-(10), where $A_n = 0$ for all n , since $f(r) = 0$. From (10) we have

$$B_n = \frac{2}{100\alpha_n a J_1^2(\alpha_n)} \int_0^1 -100 J_0(\alpha_n r) r dr = \frac{-2}{\alpha_n J_1^2(\alpha_n)} \int_0^1 J_0(\alpha_n r) r dr = \frac{-2}{\alpha_n^2 J_1(\alpha_n)}$$

Thus, from (8), we obtain the solution:

$$u(r, t) = \sum_{n=1}^{\infty} \frac{-2}{\alpha_n^2 J_1(\alpha_n)} \sin(100\alpha_n t) J_0(\alpha_n r)$$

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Solution

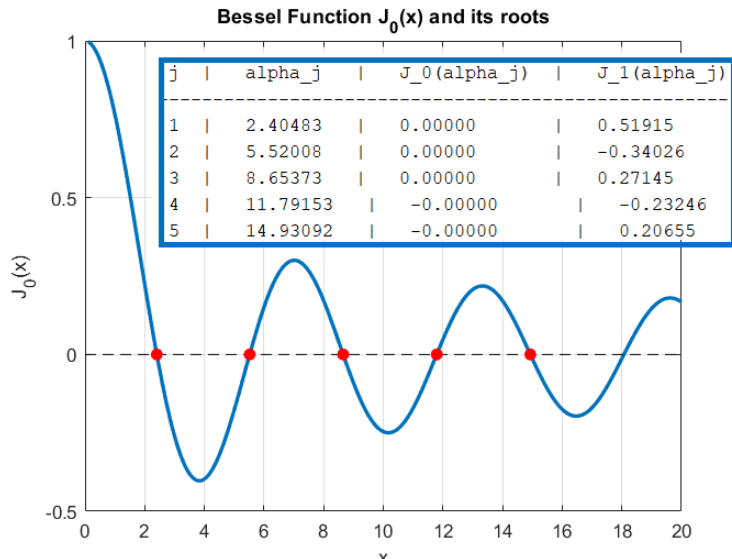
The solution is given by (8)-(10), where $A_n = 0$ for all n , since $f(r) = 0$. From (10) we have

$$B_n = \frac{2}{100\alpha_n a J_1^2(\alpha_n)} \int_0^1 -100 J_0(\alpha_n r) r dr = \frac{-2}{\alpha_n J_1^2(\alpha_n)} \int_0^1 J_0(\alpha_n r) r dr = \frac{-2}{\alpha_n^2 J_1(\alpha_n)}$$

Thus, from (8), we obtain the solution:

$$u(r, t) = \sum_{n=1}^{\infty} \frac{-2}{\alpha_n^2 J_1(\alpha_n)} \sin(100\alpha_n t) J_0(\alpha_n r)$$

Vibrations of a Circular Membrane: Symmetric Case



Vibrations of a Circular Membrane: Symmetric Case

From the table of roots, we extract the first three roots $\alpha_1, \alpha_2, \alpha_3$, and their corresponding values of $J_0(\alpha_n)$ and $J_1(\alpha_n)$:

- First root ($n = 1$):

$$\alpha_1 = 3.83171, \quad J_0(\alpha_1) = 0, \quad J_1(\alpha_1) = 0.92325$$

- Second root ($n = 2$):

$$\alpha_2 = 7.01559, \quad J_0(\alpha_2) = 0, \quad J_1(\alpha_2) = 0.62349$$

- Third root ($n = 3$):

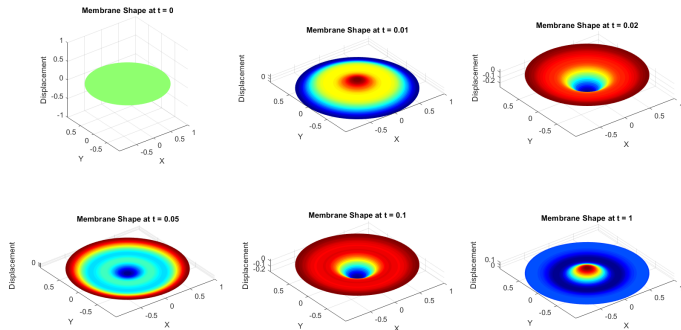
$$\alpha_3 = 10.17347, \quad J_0(\alpha_3) = 0, \quad J_1(\alpha_3) = 0.28175$$

Vibrations of a Circular Membrane: Symmetric Case

The first three terms of the solution can be written as:

$$\begin{aligned} u(r, t) \approx & \frac{-2}{14.6942 \cdot 0.92325} \sin(389.554 \cdot t) J_0(3.83171 \cdot r) + \\ & + \frac{-2}{49.2676 \cdot 0.62349} \sin(701.559 \cdot t) J_0(7.01559 \cdot r) + \\ & + \frac{-2}{103.5105 \cdot 0.28175} \sin(1017.347 \cdot t) J_0(10.17347 \cdot r) \end{aligned}$$

Vibrations of a Circular Membrane: Symmetric Case



Radially symmetric heat equation on a disk

Introduction to the Problem

- We are studying the distribution of heat in a thin circular disk with uniform thermal properties.
- The disk has its center at the origin, with a radius denoted by a .
- The heat distribution in the disk is governed by the **two-dimensional heat equation**, which describes how temperature evolves over time.
- Due to radial symmetry, we express the heat equation in polar coordinates (r, θ) , simplifying the problem by assuming the temperature r and time t .

Boundary and Initial Conditions

- **Boundary Condition:** The temperature at the boundary $r = a$ is held fixed (e.g., $u(a, t) = 0$ for a perfectly insulated or clamped edge) or follows a given function $h(t)$ (e.g., constant heat source).
- The **initial temperature distribution** is described by a function $u(r, 0) = f(r)$, which represents the temperature at time $t = 0$ for $0 \leq r \leq a$.

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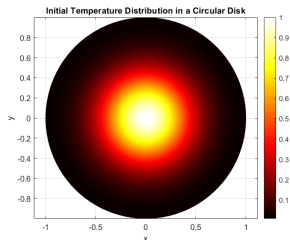
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Radially symmetric heat equation on a disk

Axisymmetric Assumption

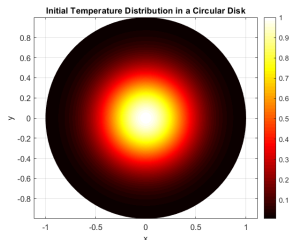
- In this problem, we focus on the case where the initial temperature distribution $f(r, \theta)$ is radially symmetric (**axisymmetric**). This means that f depends only on the radial coordinate r , and not on the angular coordinate θ .
- Similarly, if the boundary condition $u(a, t)$ is radially symmetric, it is reasonable to expect that the solution $u(r, t)$ to the heat equation will also be independent of θ .
- This assumption simplifies the two-dimensional heat equation in polar coordinates by reducing it to a one-dimensional problem in r .



Radially symmetric heat equation on a disk

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Radially symmetric heat equation on a disk

General Heat Equation in 2D Cartesian Coordinates

The heat equation in Cartesian coordinates is:

$$\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right),$$

where $u(x, y, t)$ is the temperature, x and y are spatial coordinates, t is time, and c^2 is the thermal diffusivity.

Switch to Polar Coordinates

In polar coordinates, the spatial variables are transformed as:

$$x = r \cos \theta, \quad y = r \sin \theta,$$

where $r = \sqrt{x^2 + y^2}$ is the radial distance and θ is the angle. The Laplacian operator ∇^2 in polar coordinates is:

$$\nabla^2 u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}.$$

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Radially symmetric heat equation on a disk

Assume Radial Symmetry

For a radially symmetric problem, the solution u depends only on r and t , not on the angle θ . Therefore:

$$\frac{\partial u}{\partial \theta} = 0 \quad \text{and} \quad \frac{\partial^2 u}{\partial \theta^2} = 0.$$

This simplifies the Laplacian to:

$$\nabla^2 u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r}.$$

Substitute into the Heat Equation

Substitute the simplified Laplacian into the heat equation:

$$\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right). \quad (1)$$

This is the radially symmetric heat equation.

Radially symmetric heat equation on a disk

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Radially symmetric heat equation on a disk

Boundary and Initial Conditions

To model the specific problem described:

- **Boundary Condition:** $u(a, t) = 0$ specifies that the temperature at the edge of the disk is held at zero for all $t > 0$.
- **Initial Condition:** $u(r, 0) = f(r)$ defines the initial temperature distribution at $t = 0$.

Summary of the Problem

The final equation and conditions are:

$$\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right), \quad 0 < r < a, t > 0, \quad (1)$$

with:

$$u(a, t) = 0, \quad t > 0 \quad (2)$$

$$u(r, 0) = f(r), \quad 0 \leq r < a \quad (3)$$

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Radially symmetric heat equation on a disk

Separation of Variables

Assume a solution of the form:

$$u(r, t) = R(r)T(t),$$

where $R(r)$ is a function of r only, and $T(t)$ is a function of t only.

Substitute into the Heat Equation:

$$R(r) \frac{dT}{dt} = c^2 T(t) \left(\frac{d^2 R}{dr^2} + \frac{1}{r} \frac{dR}{dr} \right).$$

Divide through by $R(r)T(t)$ to separate variables:

$$\frac{1}{T} \frac{dT}{dt} = c^2 \frac{1}{R} \left(\frac{d^2 R}{dr^2} + \frac{1}{r} \frac{dR}{dr} \right).$$

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Radially symmetric heat equation on a disk

Since the left-hand side depends only on t and the right-hand side depends only on r , both must equal a constant. Let this constant be $-\lambda^2$. We now have two equations:
For $T(t)$:

$$\frac{1}{T} \frac{dT}{dt} = -c^2 \lambda^2 \quad \Rightarrow \quad \frac{dT}{dt} + c^2 \lambda^2 T = 0.$$

For $R(r)$:

$$\frac{d^2 R}{dr^2} + \frac{1}{r} \frac{dR}{dr} + \lambda^2 R = 0.$$

Radially symmetric heat equation on a disk

Solution for $T(t)$

The equation for $T(t)$ is a first-order linear ODE:

$$\frac{dT}{dt} + c^2 \lambda^2 T = 0.$$

The solution is:

$$T(t) = e^{-c^2 \lambda^2 t}.$$

Solution for $R(r)$

The equation for $R(r)$ is a Bessel differential equation of order zero:

$$r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} + (\lambda^2 r^2) R = 0.$$

The general solution is:

$$R(r) = AJ_0(\lambda r) + BY_0(\lambda r),$$

where $J_0(\lambda r)$ and $Y_0(\lambda r)$ are the Bessel functions of the first and second kinds, respectively.

Radially symmetric heat equation on a disk

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Radially symmetric heat equation on a disk

Boundary Condition $u(a, t) = 0$

Since $u(a, t) = 0$, we apply this condition to $R(r)$:

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To ensure a finite solution at $r = 0$, we discard $Y_0(\lambda r)$ (which diverges as $r \rightarrow 0$). Thus:

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The boundary condition now gives:

$$J_0(\lambda a) = 0.$$

This implies that λ must be one of the roots of J_0 , denoted $\lambda_n = \frac{\alpha_n}{a}$, where α_n is the n -th positive zero of J_0 .

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Radially symmetric heat equation on a disk

General Solution

The solution is a sum of all modes n :

$$u(r, t) = \sum_{n=1}^{\infty} A_n e^{-c^2 \lambda_n^2 t} J_0(\lambda_n r).$$

Determination of A_n

Use the initial condition $u(r, 0) = f(r)$. At $t = 0$,

$$f(r) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r).$$

To find A_n , use the orthogonality of $J_0(\lambda_n r)$ over $[0, a]$. Multiply both sides by $r J_0(\lambda_m r)$ and integrate from 0 to a :

$$\int_0^a f(r) r J_0(\lambda_m r) dr = A_m \int_0^a r J_0^2(\lambda_m r) dr.$$

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Radially symmetric heat equation on a disk

Determination of A_n

From the properties of Bessel functions:

$$\int_0^a r J_0^2(\lambda_m r) dr = \frac{a^2}{2} J_1^2(\alpha_m).$$

Thus:

$$A_n = \frac{2}{a^2 J_1^2(\alpha_n)} \int_0^a f(r) r J_0(\lambda_n r) dr.$$

Radially symmetric heat equation on a disk

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Radially symmetric heat equation on a disk

HEAT EQUATION IN POLAR COORDINATES

The solution of the heat boundary value problem (1)-(3) is

$$u(r, t) = \sum_{n=1}^{\infty} \frac{2}{a^2 J_1^2(\alpha_n)} \left(\int_0^a f(r) r J_0(\lambda_n r) dr \right) e^{-c^2 \lambda_n^2 t} J_0(\lambda_n r),$$

where

$$A_n = \frac{2}{a^2 J_1^2(\alpha_n)} \int_0^a f(r) r J_0(\lambda_n r) dr,$$

$$\lambda_n = \frac{\alpha_n}{a}, \text{ and } \alpha_n \text{ is the } n\text{-th positive root of } J_0.$$

Radially symmetric heat equation on a disk

Example

Solve the heat problem when $f(r) = 100$, $0 < r < a$. Approximate the temperature of the hottest point on the plate at time $t = 3$, given that $a = 1$ and $c = 1$.

Solution:

With $f(r) = 100$, we have from Exercise 10,

$$A_n = \frac{200}{a^2 J_1(\alpha_n)} \int_0^a J_0\left(\frac{\alpha_n r}{a}\right) r \, dr$$

Change of variable: $\frac{\alpha_n r}{a} = s$, so that $r = \frac{as}{\alpha_n}$ and $dr = \frac{a}{\alpha_n} ds$.

Radially symmetric heat equation on a disk

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Radially symmetric heat equation on a disk

Solution:

The integral becomes:

$$A_n = \frac{200}{a^2 J_1(\alpha_n)} \cdot \frac{a}{\alpha_n} \int_0^{\alpha_n} J_0(s) s \, ds.$$

Now, the integral $\int_0^{\alpha_n} J_0(s) s \, ds = J_1(\alpha_n)$, so we get:

$$A_n = \frac{200}{\alpha_n J_1(\alpha_n)}.$$

Radially symmetric heat equation on a disk

Solution:

Hence, the solution for $u(r, t)$ is:

$$u(r, t) = \sum_{n=1}^{\infty} \frac{200}{\alpha_n J_1(\alpha_n)} J_0\left(\frac{\alpha_n r}{a}\right) e^{-\frac{c^2 \alpha_n^2 t}{a^2}}.$$

The solution represents the time evolution of the temperature of the disk, starting with an initial uniform temperature distribution of 100° and with its boundary held at 0° . The top and bottom faces of the disk are assumed to be completely insulated so that no heat flows through them.

Radially symmetric heat equation on a disk

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Radially symmetric heat equation on a disk

The maximum value of the temperature should occur at the center, since heat is being lost only along the circumference of the disk. Of all the points in the disk, the center is least affected by this loss of heat. If we take $a = 1$, $c = 1$, and $t = 3$ in the solution, we obtain:

$$u(r, 3) = 200 \sum_{n=1}^{\infty} \frac{J_0(\alpha_n r)}{n J_1(\alpha_n)} e^{-\frac{3\alpha_n^2}{a^2}}$$

To plot the solution, we need the zeros of $J_0(x)$:

$$\alpha_n = 5.5201, 8.6537, 11.7915, 14.9309, 18.0711 \dots$$

And then we approximate the solution.

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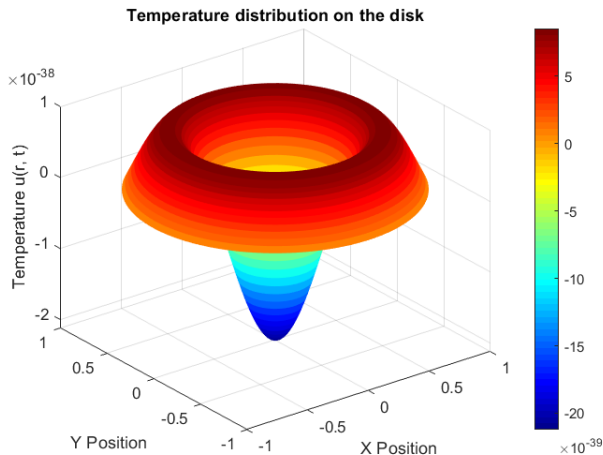
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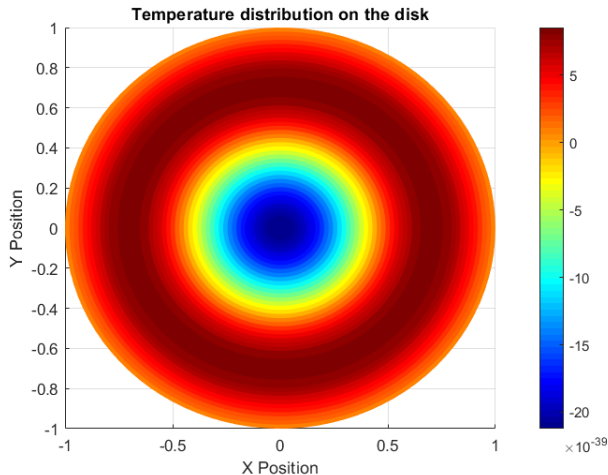
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Laplace's Equation in Circular Regions

The steady-state (or time-independent) temperature distribution in a circular plate of radius a , with prescribed temperature at the boundary, satisfies the two-dimensional Laplace equation (in polar coordinates):

$$\nabla^2 u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, \quad 0 < r < a, \quad 0 < \theta < 2\pi$$

and the boundary condition

$$u(a, \theta) = f(\theta), \quad 0 < \theta < 2\pi$$

(Note that f is necessarily 2π -periodic.)

These equations describe a **Dirichlet problem over a disk** of radius a . Following the method of separation of variables, solving the Euler equation, and imposing the boundary condition, we arrive at the following solution to this problem.

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Laplace's Equation in Circular Regions

LAPLACE'S EQUATION IN POLAR COORDINATES

The solution of Laplace's equation satisfying the prescribed boundary condition is given by

$$u(r, \theta) = a_0 + \sum_{n=1}^{\infty} \left(\frac{r}{a}\right)^n (a_n \cos(n\theta) + b_n \sin(n\theta))$$

where a_0, a_n, b_n are the Fourier coefficients of the 2π -periodic function $f(\theta)$:

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \cos(n\theta) d\theta,$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin(n\theta) d\theta$$

for $n = 1, 2, \dots$

Laplace's Equation in Circular Regions

Example: Dirichlet Problem on the Disk

Find the steady-state temperature distribution in a disk of radius 1 if the upper half of the circumference is kept at 100° and the lower half is kept at 0° .

Solution:

The boundary values are described by the function:

$$f(\theta) = \begin{cases} 100, & 0 < \theta < \pi \\ 0, & \pi < \theta < 2\pi \end{cases}$$

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Laplace's Equation in Circular Regions

Solution:

Substituting this expression for the 2π -periodic function $f(\theta)$ into the formulas for the Fourier coefficients, we get:

$$a_0 = \frac{1}{2\pi} \int_0^\pi 100 \, d\theta = 50$$

$$a_n = \frac{1}{\pi} \int_0^\pi 100 \cos(n\theta) \, d\theta = 0, \quad b_n = \frac{1}{\pi} \int_0^\pi 100 \sin(n\theta) \, d\theta = \frac{100(1 - \cos(\pi n))}{\pi n}$$

Laplace's Equation in Circular Regions

Solution:

Substituting these values into Laplace's eq., we find the solution:

$$u(r, \theta) = 50 + \frac{100}{\pi} \sum_{n=1}^{\infty} \frac{r^n}{n} (1 - \cos(\pi n)) \sin(n\theta)$$

Setting $r = 0$, we see that the temperature at the center is 50° , which corresponds to the average temperature on the boundary of the disk. On the boundary of the disk, when $r = 1$, the series becomes:

$$u(1, \theta) = 50 + \frac{200}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} \sin((2k+1)\theta)$$

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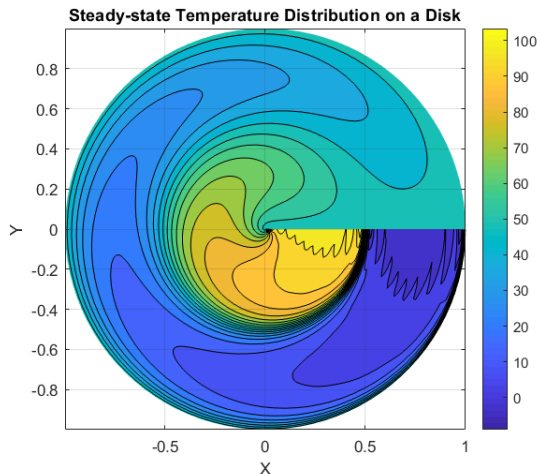


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Laplace's Equation in Circular Regions

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